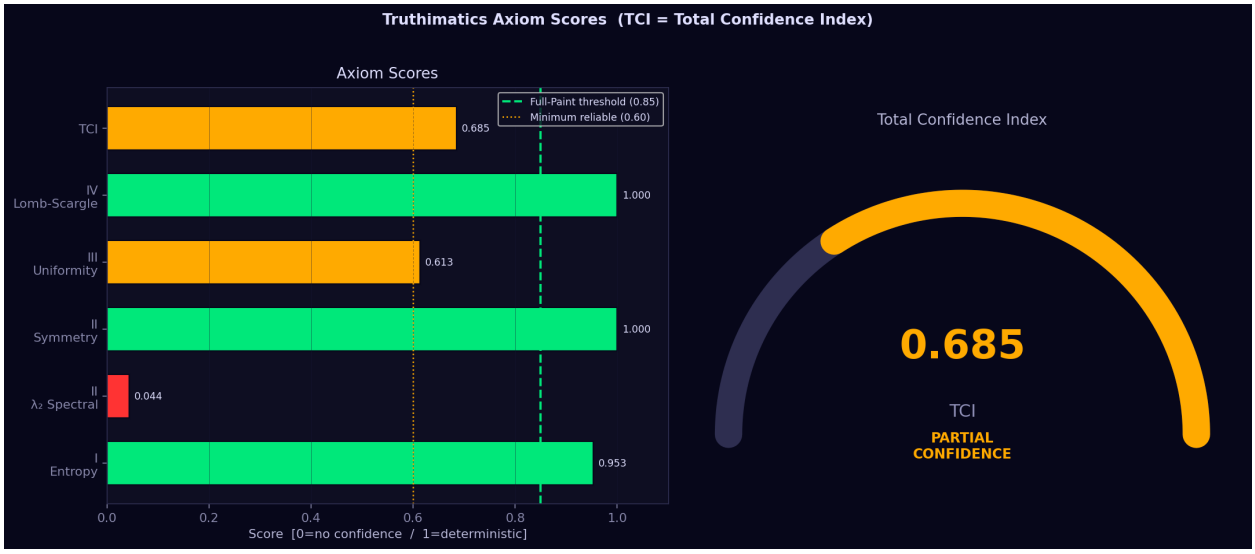
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_C59748F1B8
Taxa	Carnotaurus
Clade	Theropod
PBDB Records	1
Mass (regression)	2288.6 kg (Campione & Evans 2012)
Mass (volumetric)	2108.1 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	2.480 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6848 (partial)
Axiom I Entropy	0.9531
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.6133
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.328, 0.522, 0.0]
Generated	2026-04-30 23:52 UTC

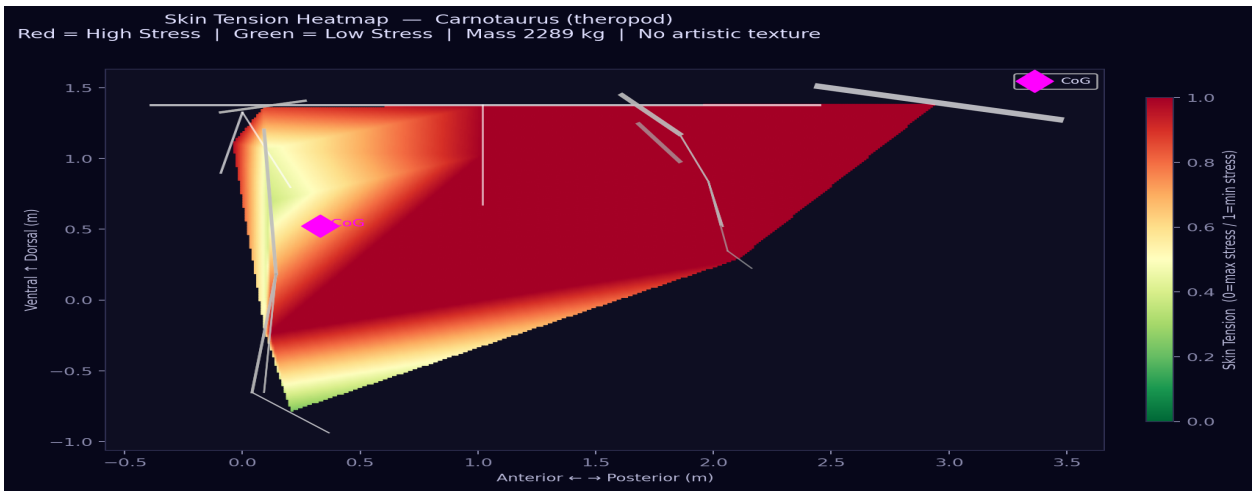
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



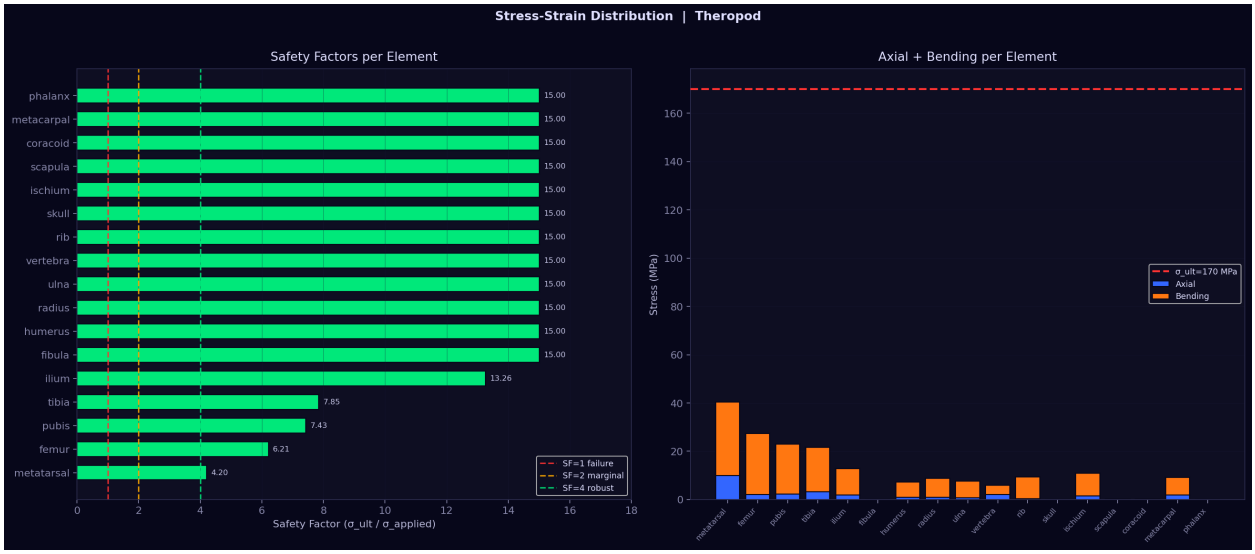
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

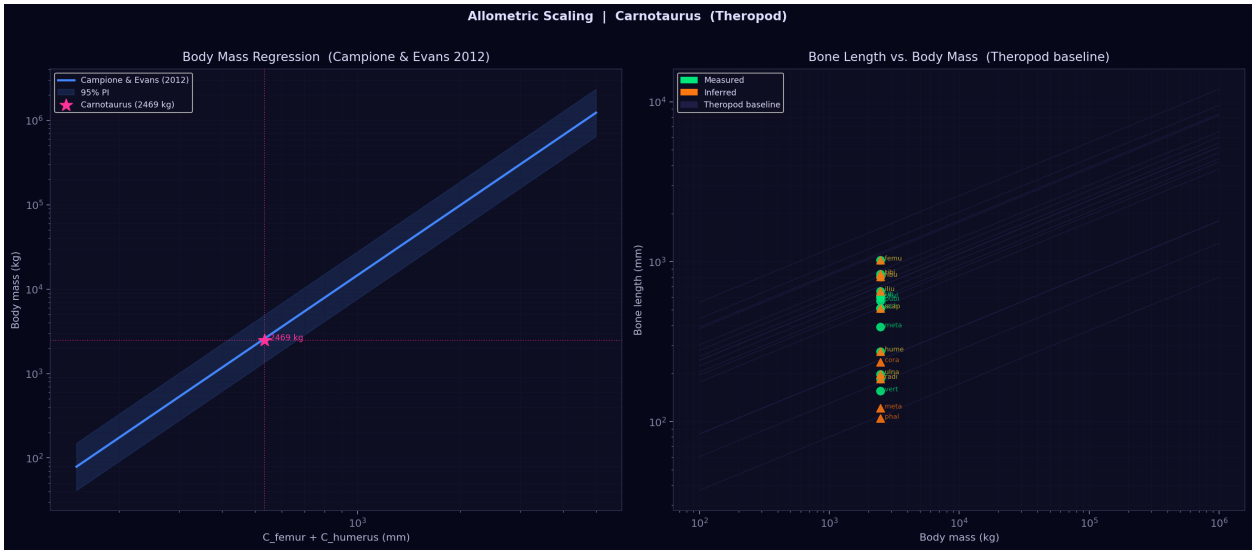


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.030	0.056	0.085	15.000	OK
femur	2.171	25.195	27.367	6.212	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.895	6.350	7.245	15.000	OK
ilium	1.935	10.888	12.823	13.257	OK
ischium	1.465	9.437	10.902	15.000	OK
metacarpal	1.899	7.348	9.247	15.000	OK
metatarsal	9.995	30.443	40.438	4.204	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	2.289	20.600	22.889	7.427	OK
radius	0.941	7.786	8.727	15.000	OK
rib	0.280	9.169	9.449	15.000	OK
scapula	0.022	0.075	0.096	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	3.281	18.385	21.666	7.846	OK
ulna	0.827	6.866	7.693	15.000	OK
vertebra	2.171	3.829	6.000	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.245 m	Femur + tibia + metatarsal
Estimated Speed	17.1 km/h (4.76 m/s)	Gait: Trot | Fr=1.029
Stride Length	5.194 m	Alexander (1976) / Thulborn (1990)
Bite Force	2345.4 N (239.1 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	595.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3634	Circumference / length (dimensionless)
Limb Strength Index	5.2e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	8.66 m	Allometric from femur length
Est. Neck Length	1.21 m	Allometric from femur length
Est. Tail Length	3.29 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	235.3	168.7	2682.84	3349.0	Constraint	0.40	15.00
femur	1020.0	118.0	5688.85	7101.4	Measured	0.88	6.21
femur_right	1020.0	118.0	5688.85	7101.4	Symmetry	0.84	6.21
fibula	805.0	40.0	364.17	381.9	Measured	0.88	15.00
fibula_right	805.0	40.0	364.17	381.9	Symmetry	0.84	15.00
humerus	275.0	52.0	297.85	371.8	Measured	0.88	15.00
humerus_right	275.0	52.0	297.85	371.8	Symmetry	0.84	15.00
ilium	655.0	125.0	4099.41	5117.3	Measured	0.88	13.26
ilium_right	655.0	125.0	4099.41	5117.3	Symmetry	0.84	13.26
ischium	510.0	85.0	1475.94	1842.4	Measured	0.88	15.00
metacarpal	121.2	34.6	21.64	17.8	Constraint	0.65	15.00
metatarsal	390.0	55.0	472.55	589.9	Measured	0.88	4.20
phalanx	104.6	35.1	15.59	12.1	Constraint	0.40	15.00
pubis	570.0	68.0	1055.73	1317.9	Measured	0.88	7.43
radius	185.0	30.0	66.69	83.3	Measured	0.88	15.00
radius_right	185.0	30.0	66.69	83.3	Symmetry	0.84	15.00
rib	610.0	30.0	219.90	274.5	Measured	0.88	15.00
scapula	510.0	198.0	8008.67	9997.2	Measured	0.88	15.00
scapula_right	510.0	198.0	8008.67	9997.2	Symmetry	0.84	15.00
skull	595.0	460.0	50430.43	62952.3	Measured	0.88	15.00
tibia	835.0	96.0	3082.40	3847.8	Measured	0.88	7.85
tibia_right	835.0	96.0	3082.40	3847.8	Symmetry	0.84	7.85
ulna	198.0	32.0	81.21	101.4	Measured	0.88	15.00
ulna_right	198.0	32.0	81.21	101.4	Symmetry	0.84	15.00
vertebra	155.0	118.0	864.48	1079.1	Measured	0.88	15.00